INCH-POUND

MIL-C-55302/42B(AT)
20 March 1990
SUPERSEDING
MIL-C-55302/42A
1 February 1973

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, ELECTRICAL, POLARIZED SHELL, FEMALE, 0.100 INCH CONTACT SPACING, REMOVABLE CRIMP CONTACTS

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-55302

This specification is approved for use by all Department and Agencies of the Department of Defense.

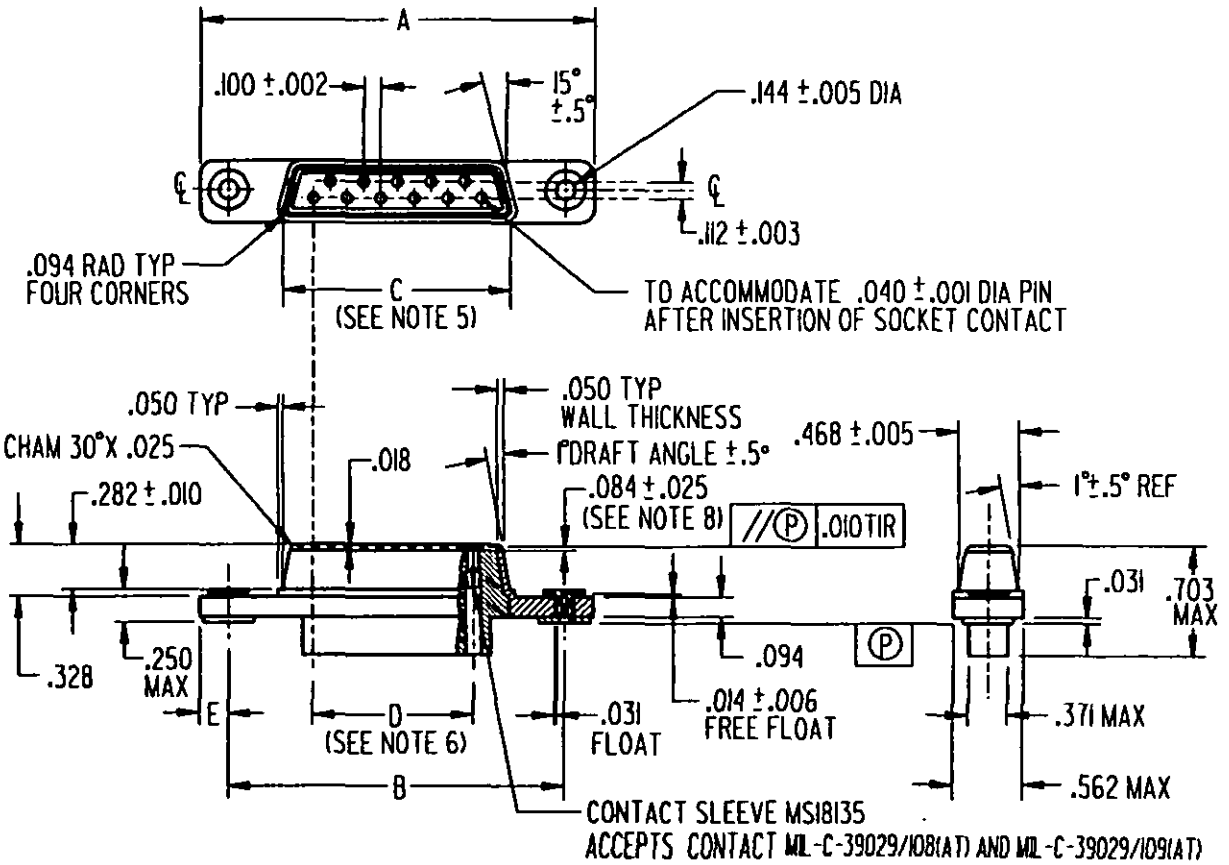


FIGURE 1. Connector, receptacle, female, .100 contact spacing, removable crimp terminal.

AMSC N/A FSC 5935

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INCHES	MM
.001	.03
.002	.05
.0025	.06
.005	.13
.006	.15
.010	.25
.014	.36
.018	.46
.025	.64
.031	.79
.040	1.02
.050	1.27
.056	1.42
.084	2.13
.094	2.39
.100	2.54
.112	2.84
.144	3.66
.250	6.35
.282	7.16
.328	8.33
.371	9.42
.468	11.89
.562	14.27
.703	17.86

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ± .015 (.38 mm).
4. Break all sharp edges .005 (.13 mm) minimum.
5. C dimension is measured at the base of the draft angle at the top of the boss, at center line of shell.
6. Nominal spacing between any two adjacent contacts shall be .100 ± .002 (2.54 ± .05 mm); overall tolerance between any span of 22 contacts shall be .100 ± .003 (.08 mm); overall tolerance between any span of 23 through 28 contacts shall be ± .004 (.10 mm); overall tolerance between any span of 29 through 35 contacts shall be ± .005 (.13 mm).
7. For dimensions A through E, see table I.
8. This dimension is from the top of casting to bottom of lead-in chamfer of socket contact MIL-C-39029/108(AT) and MIL-C-39029/109(AT).
9. No. 6 fillister head machine screw (MS35265, MS35273, MS35275 or equivalent) recommended for mounting.
10. MIL-C-39029/108(AT) or MIL-C-39029/109(AT) is to be packed separately, and shipped with the connector.
11. See M81969/1-02 for insertion and removal tool.

FIGURE 1. Connector, receptacle, female, .100 contact spacing, removable crimp terminal - Continued.

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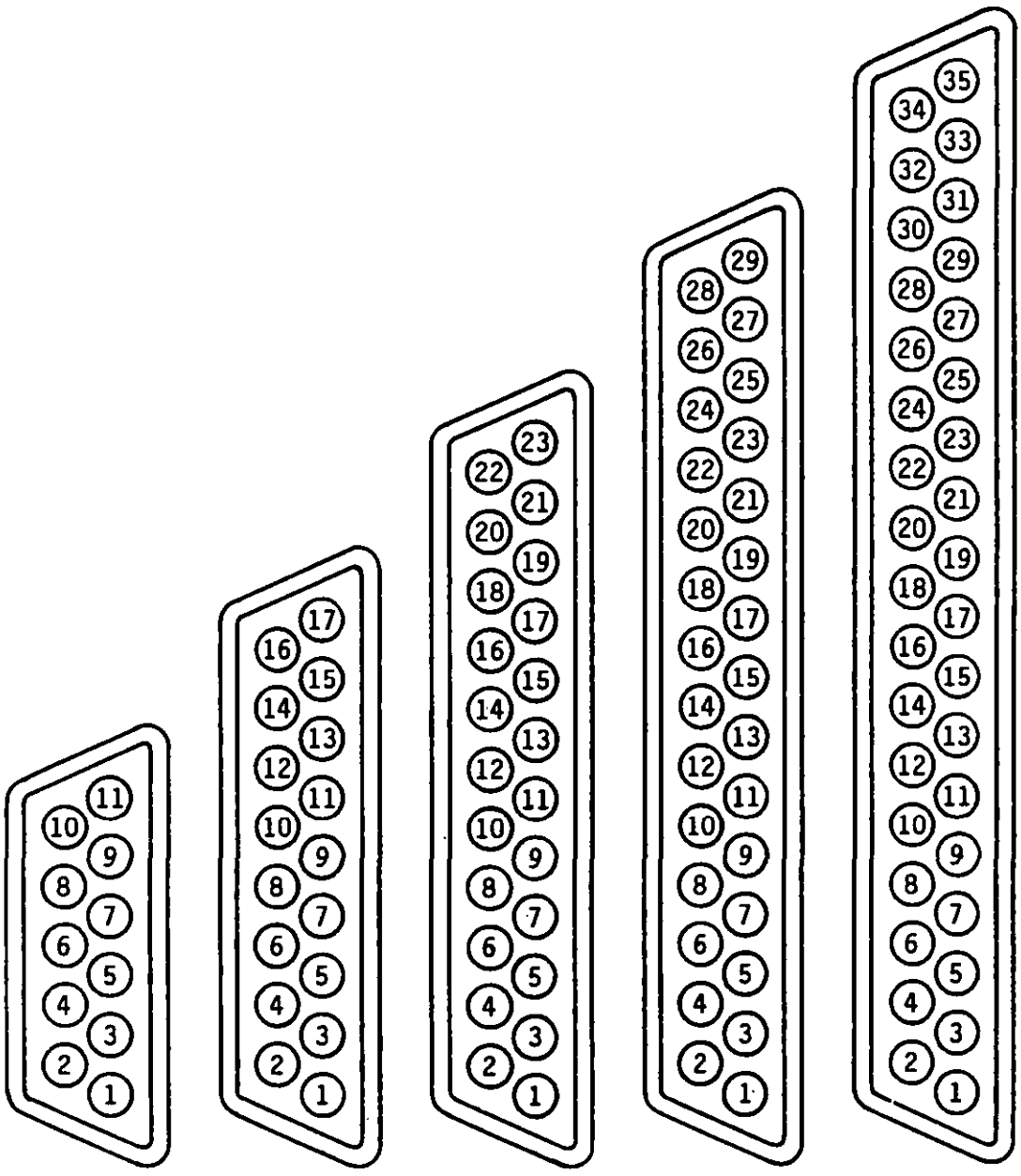


FIGURE 2. Insert arrangement for contact spacing connectors.
(Female engaging face)

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TABLE 1. Part number, dimensions and characteristics.

Part number	Dimensions 1/ 2/					Characteristics	
	A	B ± .010	C ± .005	D	E	Number of contact	Contact termination
M55302 42-01	2.594 (65.89)	2.200 (56.13-55.63)	1.244 (31.72-31.47)	1.000 (25.40)	.197 (5.00)	11	Crimp, straight
M55302 42-02	3.218 (81.74)	2.800 (71.37-70.87)	1.868 (47.57-47.32)	1.600 (40.64)	.209 (5.31)	17	
M55302 42-03	3.842 (97.59)	3.400 (86.61-86.11)	2.492 (63.42-63.17)	2.200 (55.88)	.221 (5.61)	23	
M55302 42-04	4.466 (113.44)	4.000 (101.85-101.35)	3.116 (79.27-29.02)	2.800 (71.12)	.233 (5.92)	29	
M55302 42-05	5.090 (129.29)	4.600 (117.09-116.59)	3.740 (95.12-94.87)	3.400 (86.63)	.245 (6.22)	35	

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only, and are based upon 1 inch - 25.4 mm.

TABLE II. Cross reference between MIL-C-23353 type number and MIL-C-55302 part number.

MIL-C-23353 type no.	MIL-C-55302 part no.
RC11C07	M55302 42-01
RC17C07	M55302 42-02
RC23C07	M55302 42-03
RC29C07	M55302 42-04
RC35C07	M55302 42-05

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REQUIREMENTS:

Dimensions and configuration	See figure 1 and table I.
Materials:	
Trapezoidal shells	Die cast aluminum, per QQ-A-591, compositions 380, or A380, cadmium plated, iridescent gold color, type II, class 1 or 2, or QQ-P-416.
Contact resistance	The average resistance of all contact pairs measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.
Dielectric withstanding voltage:	
Sea level	1,800 volts rms, 60 Hz. ac.
High altitude	900 volts rms, 60Hz. ac.
Working voltage	500 volts rms, maximum.
Contact current rating	7.5 amperes.
Insert arrangement	See figure 2.
Contact sleeve, MS18135, accepts contact MIL-C-39029/108(AT) and MIL-C-39029/109(AT)	Shall be included as part of the connector.
Mating connectors	See MIL-C-55302/38.
Part number	See table I.

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Army - AT

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